Dynamic Spatially-Adaptive Modulation for Image Dehazing

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Abstract

Recent works in image dehazing have demonstrated the effectiveness of spatially adaptive feature-driven operations in overcoming the inherent drawbacks of standard convolution filters. However, this approach has the following two critical disadvantages. First, training may be unstable due to degeneracy induced by hypernetwork structure and a high degree of freedom of the operations. Second, it incurs a huge extra computational burden for parameter estimation and feature computation. To tackle these limitations, we convert the pixel-wise adaptive modulation component into three disentangled—static, global, and local—pathways for hierarchical modulations, minimizing the potential degeneracy issue for learning high degree-of-freedom modulation parameters. Furthermore, we deliberately design a bottom-up branch selection algorithm to construct dynamic routes for inference depending on input features to the module without additional supervision. Our experiments demonstrate that the proposed approach effectively reduces computational cost at inference while achieving competitive accuracy on various benchmarks for image dehazing tasks. Additionally, our algorithm can be applied straightforwardly to a more general task, universal image restoration, and achieves outstanding image quality scores and recognition accuracy with a fraction of computational cost.

1. Introduction

Image restoration aims to enhance the visual quality of corrupted images formed by various sources of degradation, including motion, lighting condition, weather condition, and so on. Although existing approaches based on deep neural networks demonstrate outstanding performance on this challenging problem, it is getting more critical to learn highly adaptive models to individual examples while maintaining robustness to the datasets with significant variations. To tackle the challenges, recent image restoration methods [6, 29] have begun to adopt spatially variant operations in contrast to the conventional approaches relying on global processes such as convolutions. This is motivated by the fact that the standard convolution may not effectively maximize image restoration performance in real examples partly because there are many non-uniform distortions within an image. At the same time, the characteristics of noise or artifacts are substantially distinct across images.

The image restoration tasks with spatially varying noise, including our main target problem, image dehazing, is closely related to image style transfer [11, 15, 23, 34], which typically predicts the pixel-wise adaptive affine transform parameters to modulate normalized feature maps. However, such spatially-adaptive operations suffer from inherent drawbacks: many regressed parameters and high computational cost. Moreover, recent image restoration techniques with spatially-adaptive operations are often more vulnerable due to their degeneracy; the input feature map is employed to regress the spatially-adaptive parameters in contrast to the earlier approaches that simply rely on external inputs or components such as segmentation maps and encoders. Such a hypernetwork architecture may hamper robust training, leading to performance degradation. Despite this limitation, most existing approaches do not consider the potential stability issue for designing spatially-adaptive regressor [6, 15, 23] or are limited to employing coarse spatially-adaptive operations and reducing the number of regressed parameters [27, 29].

We are interested in incorporating spatially-adaptive modulation into image dehazing and generalizing the architecture to universal image restoration task [29, 30], which aims to restore images with various kinds of artifacts using a single model. To this end, we propose the dynamic spatially-adaptive modulation, which maximizes the stability and efficiency of our model at the same time. Contrary to the existing methods predicting pixelwise affine parameters without external inputs or components [**?**, 23], we apply the spatially-adaptive modulation components dynamically to the main network by disentangling the operation into three—static, global, and local—pathways. This strategy helps minimize potential degradation problems and speed

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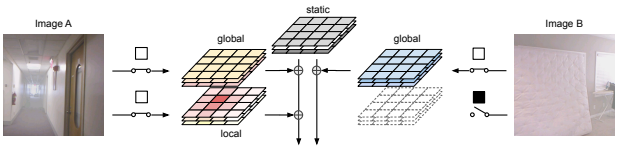

Figure 1. Conceptual diagram for our model. Our model performs spatially adaptive modulation by disentangling static, global, and local branches in a hierarchy. The scope selection module dynamically determines which branches to use using the input feature. The white and black boxes indicate the hard attentions for each branch with a value of one and zero, respectively.

up the inference process by dynamically selecting the pathways based on the characteristics of the preceding input feature maps. Fig 1 illustrates the main idea of the proposed dynamic spatially-adaptive modulation, which allows data-driven selection of the pathways.

The following is the list of our main contributions in this work:

- We propose a novel framework of the dynamic spatially-adaptive modulation for image dehazing. The proposed approach mitigates the degeneracy risk and provide good trade-off between adaptivity and robustness.
- The proposed algorithm introduces a selection module that conditionally determines the necessary modulation pathways in a bottom-up manner by optimizing accuracy and efficiency by providing a loss function.
- Our model achieves competitive accuracy in image dehazing and universal image restoration tasks with a fraction of computational cost compared to the existing methods with spatially-adaptive operations.

The rest of this paper is organized as follows. We first review prior works in Section 2, and presents our main algorithm in Section 3. Section 4 demonstrates the experimental results of the proposed approach in comparison to existing technqiues. Finally, we conclude this work in Section 5.

2. Related Work

Since we show our model's effectiveness in dehazing and extend to universal image restoration task, we will cover both topics in this section. After that, we will discuess related works for spatially adaptive affine transform and efficient inference.

2.1. Single Image Dehazing

Single image dehazing is a task to recover a clean image from a hazy input, which requires the reconstruction of missing information the input image. For example, Tan [31] uses low contrast assumption while [10] exploits dark channel prior, and [38] adopts color attenuation prior to supplement missing information.

Recently, approaches based on end-to-end trainable deep neural networks have achieved outstanding results [5–7, 18, 24, 25, 36]. Some methods [5, 25, 36] learn to explicitly estimate the transmission map and the atmospheric light as intermediate results for the *atmospheric scattering model* [22]. However, they often suffer from the propagation of the errors in the intermediate predictions and the lack of supervision for intermediate outputs. On the other hand, [18, 24, 26] train the network for capturing the latent relation between clean and hazy images without the optical model condition. GridDehazeNet [18] argues against the role of physical model in dehazing and shows that the relaxation of the physical constraint leads to performance boost with their grid architecture. HardGAN [36] extends the idea from GridDehazeNet but incorporates the mathematical model for image dehazing implicitly, where spatially-variant operations in a grid architecture is employed to maximize the flexibility of the model. We also build the overall architecture in a grid shape like HardGAN [36] and GridDehazeNet [18], and predict the output without prior information and mathematical model. Also FFA-Net [24] combines the channel attention and pixel attention providing the additional flexibility. As a result, the model shows tremendous gap from the previous works.

Besides the image dehazing task, we also present the performance of the proposed model in the universal image restoration to evaluate the generalizability of our model. Universal image restoration aims to enhance the images distorted with various corruptions in a single model. To handle such a large problem space, OWAN [30] conceived a operation selection model using soft-attention while URIE [29] used a selection model between Batch Normalization (BN) [12] and Instance Normalization (IN) [33] to solve the universal image enhancement for image recognition.

2.2. Spatially Adaptive Operations

Many visual recognition tasks benefit from spatially-adaptive operations in deep neural networks since they often need to handle the data with substantial spatial variations. Such tasks include image segmentation [32, 39], image enhancement [21, 29, 35], image-to-image translation [23,27], and so on. Spatially-dynamically operation are implemented by two kinds of techniques: spatially-adaptive modulations and dynamic convolutions.

The spatially-adaptive modulation maximizes the flexibility of the models [21, 32, 35, 39], which are typically combined with normalization operations. Although not spatially adaptive, instance normalization learns to trans-

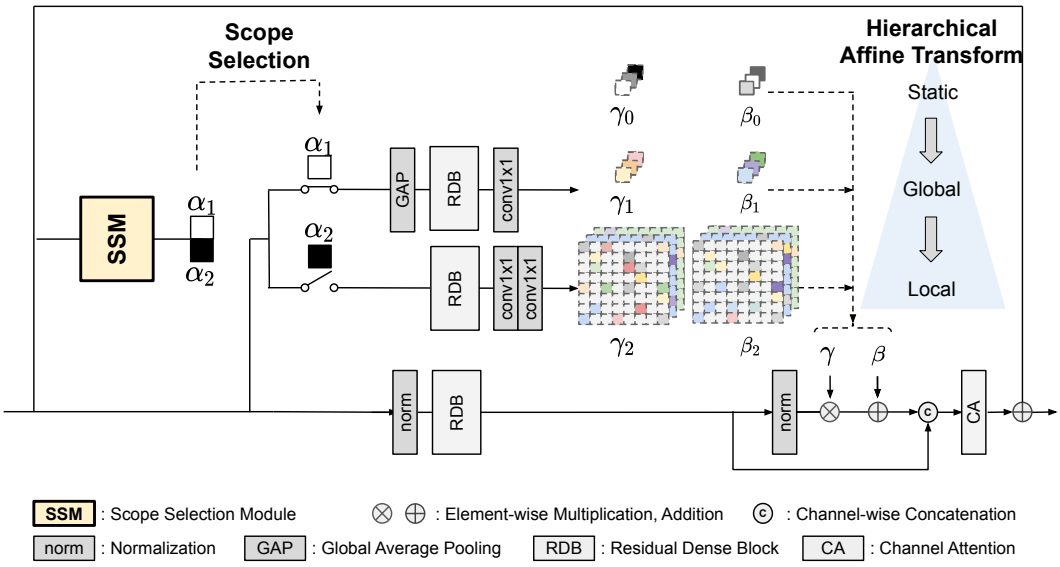

Figure 2. Our spatially-adaptive block module. We disentangle the pixel-wise adaptive modulation into three sub-branches: static, global, and local. Each branch is activated based on the hard attention from our scope selection module, SSM.

fer style across images [11, 33] while AdaIN [11] predicts channel-wise affine parameters for fast style transfer. By extending AdaIN, SPADE [23]has been proposed and applied to the spatially-adaptive denormalization module for semantic image synthesis. Kim et al. [15] and Wang et al. [34] also adopt this framework to image denoising and super-resolution, respectively.

Also, [27, 29] adopt coarse spatially-adaptive modulations by imposing a constraint on the number of regressed parameters, which are shared by neighborhood pixels to save the cost and avoid degeneracy at a cost of flexibility.

On the other hand, [14, 37] present dynamic convolution operators, which save memory due to its high flexibility traits though causing the huge computational cost at the same time. [27, 29] adopt coarse spatially-adaptive operations by imposing a constraint on the number of regressed parameters, where the regressed parameters are shared with isolated adjacent pixels to save the cost and avoid degeneracy though losing the flexibility.

3. Proposed Method

The goal of the proposed model is to restore the clean images from the hazy images under the supervision of the corresponding paired examples. By interpreting the image dehazing problem as an image style transfer task, where the haze effect is treated as a style, we perform the spatially-adaptive affine transform as in recent image-to-image translation networks [15, 23]. Contrary to the standard spatially-adaptive feature transform, which requires external input for parameter prediction, the proposed approach employs the preceding feature map to obtain the modulation parameters. Our algorithm employs a hierarchical structure for modulation and selects the optimal spatial resolution for modulation dynamically; it also reduces computational cost at inference time.

We build our overall architecture following our baselines [6] which use a grid network composed of many basic block modules. We replace their block with our novel spatially adaptive block module, described in Fig. 2. The grid-shaped network can effectively restore the hazed image with the multi-level encoder-decoder architecture, where each block is connected with its adjacent blocks by making the overall architecture looks like a grid. Please refer to Appendix for the detailed overall architecture configuration.

3.1. Hierarchical Affine Transform

Spatially adaptive feature transform As illustrated in Fig 2, our algorithm regresses the spatially adaptive affine parameters to modulate the input feature $X_i \in \mathbb{R}^{C_i \times H_i \times W_i}$ to the output feature $Y_i \in \mathbb{R}^{C_i \times H_i \times W_i}$, where C_i, H_i, W_i indicates the channel dimension, the height, and width of the feature in the i-th module, and we will omit the superscript of the block index i from now on to avoid notation clutter by simplifying X_i as X until the last of the paper. As suggested in various style transfer papers [11, 15, 23, 34], the instance normalization of activation can be regarded as the original style removal from input feature itself. The affine transform parameters have information of the new style needed for modulation and they are composed with the

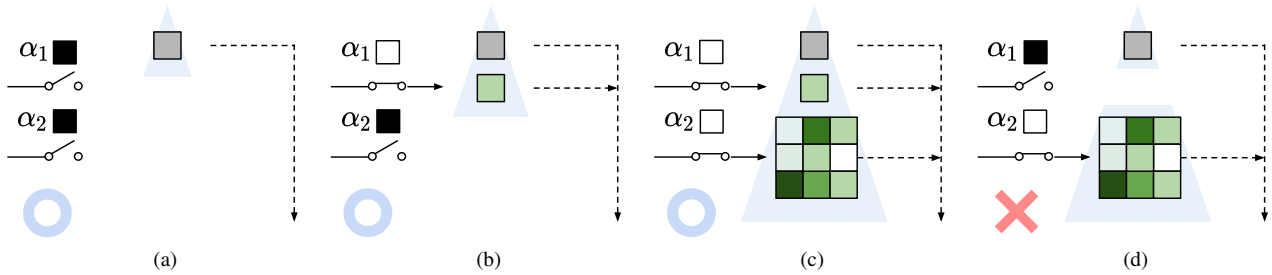

Figure 3. Selection scopes in a hierarchy. The left three diagrams show our selection scope: (a) static only, (b) static and global, (c) static, global and local branches. Note that selection scopes are defined incrementally, so that the degree-of-freedom gradually increases. The lower branch is used only when all of the upper branches are activated. The rightmost diagram (d) is not available in our setting.

scale and bias parameter $\gamma, \beta \in \mathbb{R}^{C\times H\times W}$, and modulate the instance normalized feature as follows:

$$Y = \gamma * \text{Norm}(X) + \beta. \tag{1}$$

Hierarchical modulation When we regressing the affine parameters, we disentangled the parameters into three hierarchical levels to reduce the degeneracy problem, which is because using only one hypernetwork to regress high dimensional parameter makes the operation more susceptible to the degeneracy. The final affine parameters is computed as the summation of three parameters as followings:

$$\gamma^{c,y,x} = \gamma_0^c + \gamma_1^c + \gamma_2^{c,y,x} \tag{2}$$
$$\beta^{c,y,x} = \beta_0^c + \beta_1^c + \beta_2^{c,y,x}. \tag{3}$$

The top-level components of the hierarchical structure γ_0, β_0 are learnable channel-wise affine parameters and capture overall static modulation along with the entire dataset. The mid-level components γ_1, β_1 are channel-wise adaptive parameters regressed from the residual hypernetwork branch, and aim to cover adaptive global modulation which is spatially shared in each image. The bottom-level components γ_2, β_2 are spatially adaptive parameters which are varying in the spatial domain, so they are regressed from the other residual branch without pooling.

The goal of designing the bottom component is to handle the fine local modulation in an image. However, without any regularization, the bottom modulation can invade the role of upper level modulations and the robust training can be hampered due to its flexibility. We regularized each bottom level parameter to ensure having zero-mean value along with the spatial domain, where the zero-mean property ensures the hierarchical structure between the adaptive parameters because there is no global shift-induced from local level modulation.

$$\gamma_2 \leftarrow \delta_\gamma^c * \text{Norm}\left(\gamma_2\right) \tag{4}$$
$$\beta_2 \leftarrow \delta_\beta^c * \text{Norm}\left(\beta_2\right), \tag{5}$$

where the Norm indicates the instance normalization which norms the feature in spatial domain. Then additional learnable parameters $\delta_\gamma^c, \delta_\beta^c$ modulates the variance in spatial domain of the local level modulation.

3.2. Scope Selection Module

The selection module is the central part of our model, which significantly reduces the computational cost with few performance drop by using bottom-up decision strategy for decision. Unlike the settings so far, we reduce the inference time by skipping unnecessary scope because not all of the repeated blocks would require such expensive operations. In inference time, we selected the necessary scope and the input feature proceeds only the selected scope branches, which is an effective method to regulate the use of the flexible model.

Selecting scope We define the hierarchical scope as described in Fig 3. There are three possible selection scopes: (a) static only (b) static and global (c) static, global and local. Since all branches are disentangled with the branch right above, the selection scopes are defined incrementally. Note that selecting static and local branch excluding global branch is prohibited due to the discontinuity of the degree-of-freedom.

Bottom-up decision strategy However, the asymmetry of selection scopes in a hierarchy make it impossible to determine whether or not to use each branch simply in a parallel manner. Because the upper branch is always learned more frequently, it can adversely affect the probability of choosing a lower branch. For example, one of the naive algorithms for this architecture is to select the module in a top-down manner. Because we decide which branch to use sequentially, the lower module is less likely to be updated. Even worse is that as it is better not to choose the untrained module for the performance, so the probability of using the lower module diminishes.

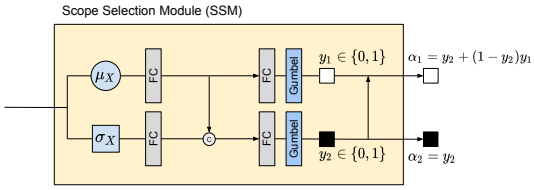

Figure 4. The scope selection module (SSM). Here, y_1, y_2 are binary logits obtained by the Gumbel-max trick. α_1, α_2 are the hard attentions which decide the final use for each branch. μ_X and σ_X stand for the mean and standard deviation of input feature X in a spatial dimension. Note that the decision for spatially-invariant operation y_1 only depends on the spatially-invariant term, μ_X, while spatially-variant term y_2 depends on both μ_X and σ_X.

To solve this issue, we newly suggest the bottom-up selection algorithm based on Gumbel-max trick [13]. Unlike other methods, we determine whether to use the bottom branch first, in our case, the local transformation. If the algorithm decides to use the bottom branch, then we use all three branches regardless of other branch use. However, if it decides not to use local branch, then we investigate the use of branch right above, global transformation.

Following these steps, the hard attentions for each branch can be expressed as follows:

$$\alpha_1 = y_2 + (1 - y_2)y_1 \tag{6}$$

$$\alpha_2 = y_2 \tag{7}$$

where $\alpha_1, \alpha_2 \in \{0, 1\}$ are the final attentions for the global and local branch use, while $y_1, y_2 \in \{0, 1\}$ are the discretized logits from Gumbel selection module, representing the use of each branch when all lower branches are deactivated. The scope selection module SSM) for this logic is described in Fig 4. Please note that the upper branch has incremental probability. For example, if $\alpha_2 = y_2 = 1$, then $\alpha_1 = 1$ regardless of y_1. This is because the lower branches are always trained with the upper branches.

Most importantly, the decision for the lower branch is not affected by the upper branch. As we mentioned earlier, more trained upper branch can inherently hinder the lower branch's training, inducing to fall into local minima. But by choosing lower branches first, selection for lower branch can be trained regardless of upper branch's performance. With these traits, we can resolve the selection of imbalance problem. Applying these attentions into Eq 4 and Eq 5 gives the final affine parameter for our spatially-adaptive block module as follows:

$$\gamma^{c,y,x} = \gamma_0^c + \alpha_1\gamma_1^c + \alpha_2\gamma_2^{c,y,x} \tag{8}$$

$$\beta^{c,y,x} = \beta_0^c + \alpha_1\beta_1^c + \alpha_2\beta_2^{c,y,x} \tag{9}$$

In conclusion, we were able to effectively reduce the computational cost by hierarchical affine transformation with a bottom-up selection algorithm.

3.3. Objective Function

Our objective function uses the smooth L1 [9], perceptual and a flops minimization loss as follows.

$$L_{total} = \lambda_1 L_1 + \lambda_{per} L_{per} + \lambda_{flops} L_{flops}, \tag{10}$$

where λ_1, λ_{per} and λ_{flops} are hyperparameters with value $1, 4 \times 10^{-2}$ and 5×10^{-7}, respectively. The smooth L1 loss and perceptual loss reduce the difference between the prediction and ground-truth images in pixel and feature level. For the perceptual loss, we use the feature distance from the first three layers of VGG-19 [28] pre-trained on Imagenet [16].

FLOPS minimization loss To suppress computationally expensive branches in every block, we directly use FLOPS loss.

$$L_{flops} = \sum_i (\alpha_1^i FLOPS_1^i + \alpha_2^i FLOPS_2^i) \tag{11}$$

Here, $FLOPS_1^i$, $FLOPS_2^i$ are the calculated FLOPS for the global and local branch in i-th block, respectively. Note that the branch with larger number of FLOPS gets bigger penalty inducing to deactivation. By controlling λ_{flops}, we can manage the trade-off between performance and computational cost by regularizing global and local branches.

4. Experiments

To show our model's efficiency, we first conduct the experiment on dehazing in two datasets, RESIDE [17] and NTIRE2018 [3, 4]. After we show the ablation study on various settings to validate suggested methods, we further extend our module to universal image restoration task showing generalizability of our module.

For more comprehensive analysis, we also conduct the experiment on our model to the universal image restoration task, we plug in our module to the previous image restoration model URIE [29] More examples are documented in the Appendix.

4.1. Dataset and Implementation Detail

We used both indoor and outdoor scenes included in RESIDE and NTIRE2018 dataset for training and test. For optimization, we utilize Adam optimizer with $\beta_1 = 0.9$ and $\beta_2 = 0.999$, and we set an initial learning rate as 0.0005, which is halved every 20 epochs. It was implemented using Pytorch and experimented with 4 Titan XP GPUs. The detail implementations slightly differ to each other depending on datasets for fair comparison with each benchmarks.

Table 1. Quantitative comparision on the benchmark RESIDE Dataset.

Method	#Params↓	#FLOPS↓	Indoor		Outdoor	
			PSNR↑	SSIM↑	PSNR↑	SSIM↑
AODNet	-	-	20.50	0.816	24.14	0.920
DehazeNet	-	-	19.82	0.821	24.75	0.930
GCANet	0.7M	18.4B	30.23	0.980	-	-
GridDehazeNet [18]	1.0M	18.9B	30.23	0.984	30.86	0.982
FFANet [24]	4.5M	253.18B	36.39	0.956	33.57	0.984
HardGAN [6]	2.6M	43.2B	36.56	0.991	34.34	0.987
Ours	3.1M	32.9B/28.07B	36.26	0.989	34.31	0.986

Table 2. Quantitative comparision on the benchmark NTIRE2018 Dataset.

Method	#Params↓	#FLOPS↓	I-HAZE		O-HAZE	
			PSNR↑	SSIM↑	PSNR↑	SSIM↑
PFFNet [20]	20.9M	7.3B	16.01	0.740	18.76	0.669
GridDehazeNet [18]	1.0M	21.0B	16.62	0.787	18.92	0.672
DuRN [19]	9.1M	110.2B	21.23	0.842	20.45	0.688
HardGAN [6]	2.5M	49.2B	18.81	0.775	24.24	0.743
MSBDN-DFF [7]	31.4M	41.5B	23.93	0.891	24.36	0.749
Ours	3.1M	34.1B/32.8B	19.53	0.781	24.24	0.742

RESIDE The RESIDE-standard dataset which contains 13990 image pairs and RESIDE-V0 dataset which contains 296695 images pairs after removing duplicated data in dataset, are used for the Indoor training dataset(ITS) and the Outdoor training dataset(OTS), respectively. This is the suggested method in [6, 18], and we use this method to use these works as a benchmark. For test data, we used Synthetic Objective Testing Set(SOTS) which consists of 500 images for both indoor and outdoor set, which are generated from 50 indoor and outdoor clean images, respectively. We set batch size 14, 20 for indoor and outdoor dataset respectively, and all images are randomly cropped by 240×240 size.

NTIRE2018 NTIRE2018 [2] is suggested as a challenge dataset which contains real world high resolution images. The datasets included both indoor and outdoor scenes, where they are called by I-HAZE and O-HAZE respectively. As following the method suggested in [7], the images are randomly cropped by 256×256 and randomly flip them horizontally or vertically before training.

DIV2K-C & PASCAL VOC2012 As in the previous universal image restoration paper URIE [29], we used 16 corruptions defined in ImageNet-C for train and test dataset. The intensity of corruption is applied from level 1 to 5, depending on severity, and all intensity levels are used for learning. The distortion algorithm defined above is applied to DIV2K dataset [1].

Furthermore, we use PASCAL VOC2012 [8] to measure improvements in segmentation results of restored images. In PASCAL VOC, we only apply corruption with severity level 3, and we did not use these images for learning following previous papers but just for evaluation. In other words, the restoration model is trained on DIV2K first and restored the mid-level distorted images of the PASCAL VOC.

4.2. Performance evaluation

Dehazing Our results clearly show the computational cost drop with still noticeable results in both RESDIE and NTIRE2018 dataset. Table 1 shows the quantitative results on RESIDE indoor and outdoor settings. As we expected, by regularizing the flops to optimize the model, we could hugely drop the excessive computation with noticeable margin. In outdoor dataset, almost 35% of FLOPS decreases compared with our baseline HardGAN , still remaining the overall performance. Table 2 shows the result on the NTIRE2018 dataset. Among the models with comparable results, our model shows rational performanc trading between computational cost and performance. For the qualitative results comparsion, please refer to Appendix.

Universal image restoration We compare with the state-of-the-art universal image restoration models, URIE. Table. 3 shows the quantitative result on DIV2K dataset. It shows that our model is all ahead of URIE-MSE in PSNR,

Table 3. Quantitative results of URIE and ours on DIV2K dataset. #P denotes the number of parameters. We trained both networks using MSE loss and evaluated on the same dataset, DIV2K, and all restoration scores exceed the baseline URIE.

	#Params ↓	#FLOPS↓	PSNR	SSIM	MS-SSIM
URIE [29]	671K	3.07B	20.74	0.629	0.828
Ours (w/o flop loss)	266K	3.12B	21.66	0.670	0.845
Ours	266K	**2.48B**	**21.20**	**0.656**	**0.834**

Figure 5. Qualitative results of URIE and ours on DIV2K (left) and PASCAL VOC (right) datasets. From left to right: corrupted image, URIE [29], ours, ground truth. The below image in PASCAL-VOC images is a segmentation output from the above image using pre-trained DeepLabv3. Note that the performance of segmentation as well as restored images has significantly improved. Best viewed in zoom.

SSIM, and MS-SSIM. Note that the size and the number of flops of ours is smaller than the existing method and outperforms with the noticeable margin. We could also control the tradeoff between computational cost and performance regulating the flops loss. Comparing with URIE, our model clearly shows much clearer result handling spatially-variant distortion much better. All qualitative results will be documented and discussed in Appendix.

4.3. Ablation study

To validate the novelty of our algorithm, we exahaustively conduct the ablation study on RESDIE inddor dataset. Table 5 summarizes the ablation study results. In the table, zero mean state the use of local branch normalization, and λ_{flops} indicate the hyperparameter for computational cost. Here ✓is the default value with $5 * 10^{-7}$. And std shows the use of standard deviation for the SSM.

Hierarchical Modulation In this section, we perform model structure ablation regulating the degree of freedom of our model.

Since asymmetry in a hierarchical structure, the upper branch is always trained more than the lower branch. Unlike our method, this asymmetry negatively affects the selection of the lower branches in top-down and parallel approaches, leading to unstable training in the end.

Effectiveness of adaptive FLOPS reduction algorithm To demonstrate the cost-effectiveness of our adaptive algorithm, we compare our model with other static models. We make the comparing models by regulating *growth rate* in RDB, the number of dense layers in RDB, and the width of grid architecture of the original model to reduce the FLOPS, where the detail of RDB (residual dense model) and grid architecture are explained in the last section of our supplementary.

We train each model on the various range of FLOPS from scratch and compare the performances. Unlike these static methods, our algorithms adaptively reduce FLOPS by increasing the weight of flops loss, and we plot PSNR, SSIM graphs against the number of FLOPS in Figure **??**. While reducing 35% of computational cost, our model shows only a minor performance drop (0.43dB), whereas others' performances are significantly decreased. These results show that our design is highly effective for reducing the computational burden, which is because that our algorithm dynamically optimizes the inference path while others could not effectively handle the distortion with smaller model size.

5. Conclusion

We propose a novel hierarchical spatially adaptive with a modulation selection module to solve this problem. The proposed spatially-adaptive module defines affine transforms in a three-level hierarchy for stability and captures static, global, and local modulation. To handle the various level of degree-of-freedom in a single module, we incrementally define the selection scope for optimization. Furthermore, by adopting a bottom-up selection approach, we could ensures that all branches in the hierarchy get a fair opportunity to update their weight and stabilize the overall selection process. As a result, our experiments have shown a significant computational cost drop without losing performance on both indoor and outdoor scenes in the RESIDE dataset. Additionally, we figure out our algorithm can also be applied to a more generic task, universal image restoration, and achieves the state-of-the-art performance in terms

Table 4. Ablation study of our model. All the methods are evaluated on the RESIDE indoor dataset using the same setting as the proposed algorithm except for the conditions mentioned.

Method	hierarchical	zero mean	λ_{flops}	std	PSNR↑	SSIM↑	#FLOPS↓
Non hierarchical	-	-	-	-	35.20	0.9878	26.3B
⓪	✓	-	-	-	36.20	0.9895	45.5B
①	✓	✓	-	-	36.45	0.9899	45.5B
②	✓	✓	✓	-	35.48	0.9879	34.9B
③	✓	-	✓	✓	35.83	0.9892	32.9B
Ours-v1	✓	✓	✓	✓	36.26	0.9893	32.9B
Ours-v2	✓	✓	10^{-6}	✓	36.02	0.9889	29.7B

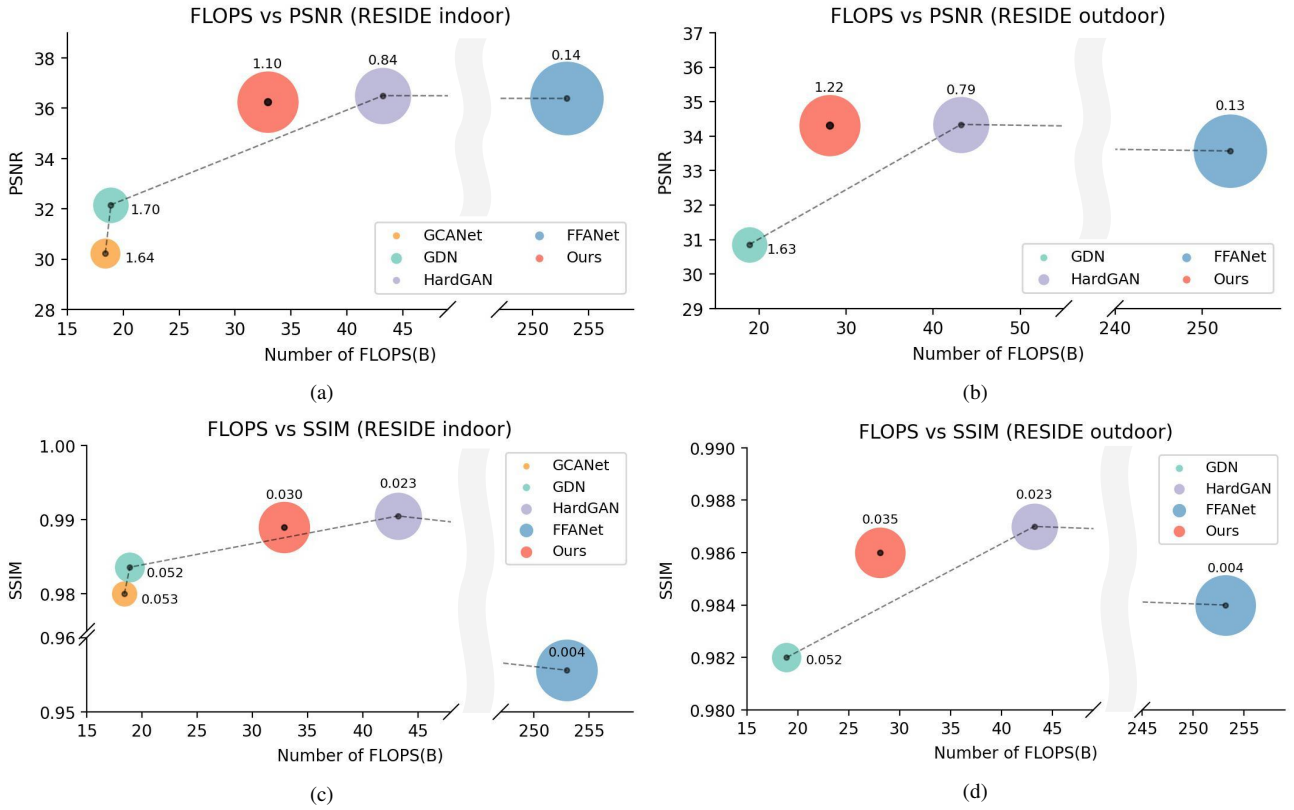

Figure 6. PSNR[dB] and SSIM performance against FLOPS, where RESIDE and SOTS are used for train and evaluation set, respectively. The size of circle represents the number of parameters, and the number written on each circle is PSNR/FLOPS or SSIM/FLOPS. The gray dotted line indicates the flops-performance trade-off line among previous methods. The proposed method, whose color is red, is located to the left and above the trade-off line, which indicates that it is superior in performance compared to FLOPS.

Table 5. PSNR[dB] and SSIM on RESIDE indoor dataset with different scope selection algorithms. It shows our bottom-up selection strategy outperforms the top-down and parallel selecting.

Method	PSNR↑	SSIM↑
Top-down	35.60	0.9884
Parallel	35.92	0.9882
Bottom-up (Ours)	**36.26**	**0.9893**

of image quality scores and recognition accuracy with the lowest computational cost, but it sometimes sensitive for finding optimal trade-off points. We expect our module can boost the performance on any in-the-wild vision tasks in the light of pre-processing, but it sometimes effect negatively on environments because using of lots of resources.

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